

CST Third Semester Midterm Examination, November 2022
Signals and Systems (CS-2104)

Answer Q1 and any three questions from the rest.

Weightage of Q1 is double that of the rest.

Partial or incomplete answer to Q1 may be supplemented by answering all of the rest.

Full Marks: 30

Time: 120 minutes

1. A second order discrete system is described by the difference equation:
 $y[n] - 0.5y[n-1] + 0.04y[n-2] = 3x[n]$.
 - (a) Obtain the transfer function $H(e^{j\omega})$ of the system.
 - (b) Obtain the output sequence expression for $x[n] = 2\delta[n] + \delta[n-1]$ as input.
Hint: Apply inverse of DFT on $H(e^{j\omega})$ to get to $h[n]$ and then apply convolution $y[n] = x[n] * h[n]$ for the given impulse expression $x[n]$.
2. A periodic square wave signal with peak to peak amplitude of 2 volts and time period of 100 ms is passed through a system that has a proportional constant of 0.5 and an integral constant of $0.01(ms)^{-1}$.
 - (a) Derive an expression for the output signal as a function of time.
 - (b) Sketch the output signal waveform in time domain for one cycle, clearly indicating the amplitudes at few equidistant time points in one cycle.
3. Explain following Fourier transform properties with proper derivation:
 - (a) Time scaling
 - (b) Time shifting
4. Derive an expression for energy content in a signal having period of T . What is the energy content in a discrete signal in terms of its sequence given in time domain and in terms of its equivalent frequency domain?
5. Write short notes on the following:
 - (a) Trade-off between time domain and frequency domain response of filter design
 - (b) Linearity and time invariance of systems